

ESAM 420 Notes

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1 Motivating Examples

Suppose we want to launch a projectile vertically into the air and we want to create a model. We are launching it vertically so we do not have to worry about vectors. We start with Newton's Law of Gravity so we can say that

$$\frac{d^2 r}{dt^2} = -G \frac{M_e}{r^2}$$

where r is the distance from the center of the Earth, G is the gravitational constant, and M_e is the mass of the Earth. However, we want to know how far from the surface of the Earth our projectile goes so we define $r = h + R$ where R is the radius of the Earth and h is the height of the projectile relative to the ground. Since $dr = dh$, we have that

$$\frac{d^2 h}{dt^2} = -G \frac{M_e}{(h + R)^2}$$

Recalling that $g = \frac{GM_e}{R^2}$, we rewrite the equation as

$$\frac{d^2 h}{dt^2} = -\frac{gR^2}{(h + R)^2}$$

This equation has no analytical solution. However, if we make the approximation $h \ll R$, we can obtain an *asymptotic solution*.

$$\frac{d^2 h}{dt^2} = -g$$

which if we set the initial conditions to be $h(0) = 0, h'(0) = v_0$, we have that the solution is

$$h(t) = v_0 t - \frac{1}{2} g t^2$$

The problem with this is we have no semi-“precise” way to justify when we can make this assumption. We also cannot add any correcting factors. We usually start these by non-dimensionalizing. This allows us to zoom to the scale we want to work with and justify what is “small.” For the gravity problem, let us define the characteristic length and times to be

$$L_c = \frac{v_0^2}{g} \quad T_c = \frac{v_0}{g}$$

We also define some new normalized variables relative to these characteristic norms (I don't want to use variables and they're mainly about magnitudes, hence norms)

$$x = \frac{h}{L_c} \quad \tau = \frac{t}{T_c}$$

We can transform $\frac{d^2 h}{dt^2} \rightarrow \frac{d^2 x}{d\tau^2}$ via the chain rule

$$\begin{aligned}\frac{dx}{d\tau} &= \frac{dx}{dh} \frac{dh}{dt} \frac{dt}{d\tau} = \frac{1}{L_c} \frac{dh}{dt} \frac{T_c}{1} \\ \frac{dh}{dt} &= \frac{L_c}{T_c} \frac{dx}{d\tau} \\ \Rightarrow \frac{d^2 h}{dt^2} &= \frac{d}{dt} \left(\frac{L_c}{T_c} \frac{dx}{d\tau} \right) = \frac{d}{dt} \frac{dt}{d\tau} \left(\frac{L_c}{T_c} \frac{dx}{d\tau} \right) = \frac{L_c}{T_c^2} \frac{d^2 x}{d\tau^2}\end{aligned}$$

So our equation now becomes

$$\begin{aligned}\frac{L_c}{T_c^2} \frac{d^2 x}{d\tau^2} &= -\frac{gR^2}{(xL_c + R)^2} = -\frac{gR^2}{R^2(x\frac{L_c}{R} + 1)^2} \\ \frac{d^2 x}{d\tau^2} &= -\frac{1}{(x\varepsilon + 1)^2} \quad \varepsilon = \frac{L_c}{R} = \frac{v_0^2}{gR}\end{aligned}$$

We then guess our solution is some sort of expansion based off this small parameter of the form

$$x(t) = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

Our small parameter being the ratio of the characteristic length to the radius of the Earth shows us exactly when we can make certain assumptions.

Another application is the Navier-Stokes equations.

$$\begin{aligned}\nabla \cdot \vec{u} &= 0 \text{ (incompressibility equation)} \\ \rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) &= -\nabla p + \mu \nabla^2 \vec{u} \text{ (equation of motion)}\end{aligned}$$

If we non-sensationalize such that $(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{L}(x, y, z)$, $\tilde{u} = \frac{\vec{u}}{u_c}$, $\tilde{p} = \frac{p}{p_c}$, $\tilde{t} = \frac{t}{T_c}$, $T_c = \frac{L_c}{u_c}$. This transforms our equation into

$$\frac{\rho u_c}{T_c} \frac{\partial \tilde{u}}{\partial \tilde{t}} + \frac{\rho u_c^2}{L_c} (\tilde{u} \cdot \tilde{\nabla}) \tilde{u} = -\frac{p_c}{L_c} \tilde{\nabla} \tilde{p} + \frac{\mu U_c}{L_c^2} \tilde{\nabla}^2 \tilde{u}$$

Notice that the first two coefficients of the left hand side are equal and so we can rewrite the left hand side as one differential operator $\frac{D\tilde{u}}{D\tilde{t}}$. Doing some algebra we get that

$$\text{Re} \frac{D\tilde{u}}{D\tilde{t}} = -\frac{p_c L_c}{\mu u_c} \tilde{\nabla} \tilde{p} + \tilde{\nabla}^2 \tilde{u}$$

where Re is the Reynold's number which is the ratio of inertial versus viscous forces in a flow.

$$\text{Re} = \frac{\rho u_c L_c}{\mu}$$

If we take the limit as the Reynold's number go to zero, we go into the length scale of bacteria typically. If we have $\frac{p_c L_c}{\mu u_c} = 1$, we then have that the characteristic pressure is $p_c = \frac{\mu u_c}{L_c}$. Rearranging the equation we get

$$\nabla p = \mu \nabla^2 \vec{u}$$

which is known as the Stoke's limit. If instead we take the limit as the Reynold's number goes to infinity and if we set the characteristic pressure to be $p_c = \rho u_c^2$, we get

$$\frac{D\vec{u}}{D\tilde{t}} = -\nabla p + \frac{1}{\text{Re}} \tilde{\nabla}^2 \tilde{u}$$

The Laplacian term goes to zero and we have the Euler Equation for inviscid flow

$$\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) = -\nabla p$$

From this we can get d'Alembert's paradox which states that the force of drag is zero on a body moving at a constant velocity in an inviscid fluid contrary to observations. Due to asymptotics, we can add correction terms to resolve this paradox.

2 Algebraic Equations

Suppose we want to solve the algebraic equation

$$x^2 + 2\varepsilon x - 1 = 0 \quad \varepsilon \ll 1$$

Analytically we can solve this to get that $x_{1,2} = -\varepsilon \pm \sqrt{1 + \varepsilon^2}$. If we can't analytically solve, we can use two methods to approximate: expansion or iteration. Let us start with the expansion method first.

2.1 Expansion Method

Let us start with the zeroth-order approximation and set $\varepsilon = 0$. We get that our solutions are $x_{0,2} = \pm 1$. We then make the guess that our solution can be expressed as the power series

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

We then plug that back into our equation and match together powers of ε up until the powers of epsilon that we desire. Doing so gets us

$$\begin{aligned} (x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots)^2 + 2\varepsilon(x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots) - 1 &= 0 \\ 1 + 2\varepsilon x_1 + 2\varepsilon^2 x_2 + \varepsilon^2 x_1 + \dots + 2\varepsilon + 2\varepsilon^2 x_1 + \dots - 1 &= 0 \end{aligned}$$

Grouping them together by powers of epsilon gives

$$\varepsilon(2x_1 + 2) + \varepsilon^2(2x_2 + x_1^2 + 2x_1) = 0 \quad \forall \varepsilon$$

Since this equation must be true for all ε , we have that the parenthesis terms must be 0 and so we solve for x_1 and then plug that into the ε^2 expression to solve for x_2 .

$$\begin{aligned} \varepsilon : 2x_1 + 2 &= 0 \rightarrow x_1 = -1 \\ \varepsilon^2 : 2x_2 + 1 - 2 &= 0 \rightarrow x_2 = \frac{1}{2} \end{aligned}$$

Our three-term expansion is therefore

$$x = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

If we take the binomial expansion of our analytical solution, we find the same thing.

2.2 Iteration Method

We generally solve for the highest order of x

$$x^2 = 1 - 2\varepsilon x \rightarrow x = \pm\sqrt{1 - 2\varepsilon x}$$

The iteration method says that

$$x_{n+1} = \pm\sqrt{1 - 2\varepsilon x_n}$$

We start with the zeroth-order approximation

$$\begin{aligned} \varepsilon = 0 &\rightarrow x_0 = \pm 1 \\ \varepsilon^1 \quad x_1 &= \sqrt{1 - 2\varepsilon} = 1 - \varepsilon \\ \varepsilon^2 \quad x_2 &= \sqrt{1 - 2(1 - \varepsilon)\varepsilon} = 1 - \varepsilon(1 - \varepsilon) - \frac{1}{8}(2\varepsilon(1 - \varepsilon))^2 \end{aligned}$$

Cleaning all of this up we get the same thing as the previous methods

$$x_3 = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

The expansion and iteration methods are good when the zeroth-order approximation doesn't change the nature of the equation (e.g., if we have $\varepsilon x^2 + x + 1 = 0$, the zeroth-order approximation would change the equation from a quadratic to a linear equation).